

Demand for LPG in rural and urban communities

Policy insights from randomized controlled trials

Memo prepared by the Combating Household Air Pollution (CHAP) project¹

Since 2020, researchers from the CHAP project conducted two trials to test the effect of price and convenience (distance) on demand for liquid petroleum gas among rural and urban households in Ghana. The market conditions in the trial were designed to approximate purchasing options under the cylinder recirculation model (CRM). The study generates four major policy implications as outlined in this brief:

1. Demand is sensitive to price, with greater sensitivities in rural areas
2. In rural areas, when prices are low distance to the LPG exchange point affects demand.
3. At anything close to market prices, LPG take up by rural communities is effectively zero.
4. Past experience with LPG helps boost demand.

Background and study design

In order to investigate potential determinants of household-level LPG demand, we conducted two separate studies: one focused on rural areas and one on urban areas. In the rural study, conducted throughout 2021, we offered study participants a variety of prices and distances to access 14.5 LPG cylinder exchanges. During the urban study, held between April and October 2022, we offered participants a variety of prices for 3 or 14.5 LPG cylinder exchanges and distances. Prices offered to participants in the rural study were much lower per kilogram (kg) (up to 5 cedis per kg), while distances were much further; in the urban study, prices per kg ranged much higher (up to 10 cedis per kg),

¹ The CHAP project is a collaboration between researchers at the Kintampo Health Research Centre, Columbia University and the University of California. Please contact Teresa Tawiah (theresa.tawiah@kintampo-hrc.org) or Kelsey Jack (kelseyjack@berkeley.edu) with questions. The memo was prepared by Kelsey Jack, Teresa Tawiah, Darby Jack, Paulina Oliva, Georgette Owusu-Amankwah, Linnea Graham, and Annelise Gill-Wiehl.

while distances to exchange LPG were much closer (Table 1). We report more specific study details and the corresponding results and policy implications below.

Table 1. Rural and urban study overview and comparison

	Rural	Urban
Time of study	January 2021-December 2021	April to October 2022
Location	North and South Districts of Kintampo	Techiman
Sample size	886	543
Prices per kg offered	0 to 5 cedis per kg	0 to 10 cedis per kg
Cylinder size offered	14.5kg	3 or 14.5 kgs
High/low LPG price distinction	High: 3.37 to 4.89 cedis per kg Low: 0 to 0.28 cedis per kg	High: equal to or over 8 cedis per kg Low: equal to or under 4 cedis per kg
LPG price per kg	Roughly 5 cedis per kg	Roughly 11 cedis per kg (April 2022)
Distance	0 to 80.6 km	0.152 to 2.3 km

Rural study

Throughout 2021, the rural study was conducted in communities in Kintampo North and South Districts in the Bono East Region of Ghana (see Figure 1). This is a predominantly rural area with a population of about 228,634. The study sampled 886 female adults from 36 communities.² All study participants were sampled from prior NIH-funded trials on exposure to household air pollution from cooking sources.³

² Two additional communities were used to pilot the study design and data collection instruments.

³ Two previous trials are referred to as GRAPHS and ELAG, and provided study participants with access to subsidized LPG in a cluster randomized trial design. We leverage the variation in past experience with LPG from these prior interventions in our analysis.

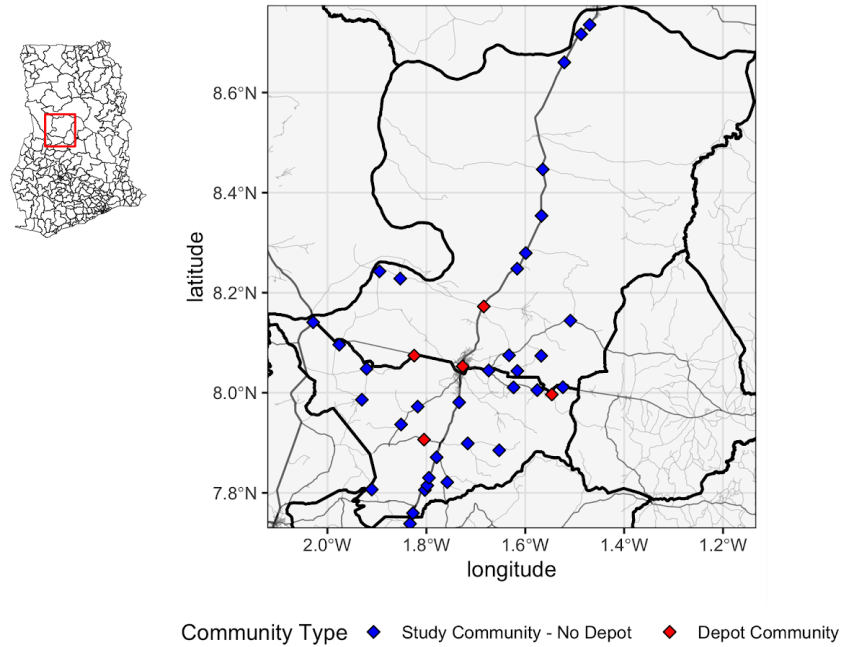


Figure 1: Map of the rural study area

Notes: Rural study area in Kintampo North and South Districts. The study communities are indicated with diamonds. Blue diamonds represent communities without an LPG exchange depot. Red diamonds indicate the location of LPG exchange depots. Sources: Main map uses data from OpenStreetMap and insert map uses data from HDX (data.humdata.org/CC).

Prices were varied at the individual level and included five price points – 0, 0.28, 1.79, 3.38, and 4.90 per kg– which represent a subsidy of approximately 100, 95, 65, 35 and 5 percent subsidy on the market price at the time of the study, respectively. Participants drew their subsidy from a bucket, without knowing the distribution of prices. The subsidy could be applied to up to 2 exchanges/refills of a 14.5 kg cylinder of LPG per month for the six months of the study.

Table 2: Rural study and participant summary statistics

	Mean	SD	Min	Max
LPG price	29.86	26.56	0.00	71.00
Distance to depot	24.61	19.68	0.00	80.61
Participant age	35.00	7.33	20.0	55.00
Participant can write	0.24	0.43	0.00	1.00
Household size	8.12	4.44	1.00	33.00
Has cash income	0.82	0.38	0.00	1.00
Monthly cash income	631.62	897.04	5.00	6000.00
Primary stove: three stone	0.91	0.29	0.00	1.00
Primary fuel: firewood	0.92	0.27	0.00	1.00
Use LPG in past 3 months	0.10	0.30	0.00	1.00

Notes: Summary statistics for main rural study treatments (price and distance) and study participants.

Distance was varied at the sub-community level and proceeded in two steps. First, communities were randomly assigned to two of five LPG exchange or refill locations (four of the former and one of the latter). The refill station was located in one of the main towns in the region, Kintampo, and was the status quo location for refilling LPG cylinders. The four exchange depots were established for the purposes of the study. These were located in randomly selected communities, chosen based on centrality, level of market activity, and location relative to Kintampo. All households in a community that received an exchange depot were assigned to that location. Otherwise, assignments were random and stratified on previous exposure to LPG, and whether travel to the depot required changing buses in Kintampo (study quadrant). Participants then randomly drew one of the two depots assigned to their community.

The random price and distance variation, along with characteristics of study participants are shown in Table 1. On average, participants are of low socioeconomic status: only 25 percent can write, and while over 80 percent have some cash income, mean household income is only around 630 cedis, for a mean household size of 8. Most households (over 90 percent) use a three stone fire as their primary cook stove and burn firewood as their primary fuel. Though all households own an LPG stove and cylinder, only 10 percent report having used it in the last three months.

Urban study

The urban study was conducted between April and October 2022 in Techiman, a city of roughly 250,000 people also in the Bono East Region of Ghana (see Figure 2). All study participants (N=543) were sampled from the Health and Demographic Surveillance System census data run by the Kintampo Public Health Research Centre.

In the first stage, respondents paid a deposit for an LPG starter kit (stove and 3kg or 14.5kg cylinder), understanding that after the 3 month study they could keep their kit, get their money back or switch cylinder sizes. Each respondent then participated in three multiple price lists to reveal their willingness to pay for 3 or 14.5 kg LPG cylinder refills but were then randomly assigned an LPG subsidy amount and a depot (first closest or second closest) that they could exchange their cylinder over the study period (Figure 2). Participants were offered prices of 0.33 to 10.07 cedis per kilogram. Distance to the depot ranged from 152 meters to 2.3 kilometers; thus, urban users face notably shorter distances to exchange than those in our rural study.

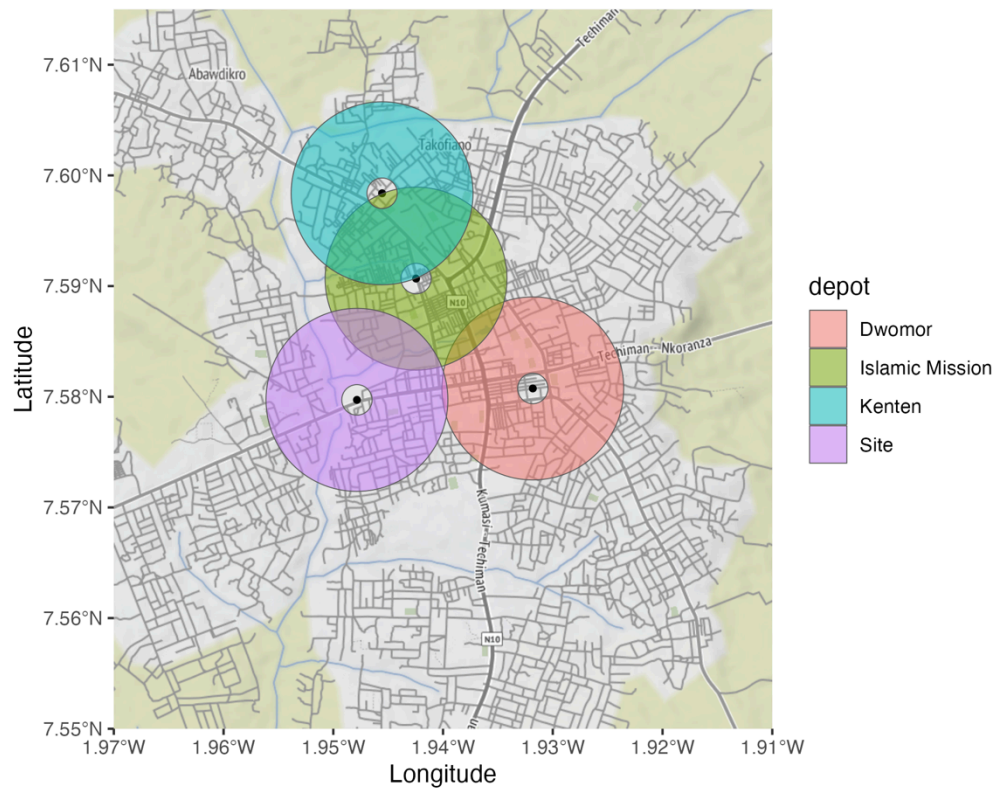


Figure 2: Map of LPG exchange locations in Techiman (urban study)

Notes: Study area in Techiman. The LPG exchange depot locations in the urban study are indicated with dots. Colored radii indicate the general catchment area for which it would be the closest depot for study participants. Participants were assigned to either the closest or the second closest depot.

Sources: Main map uses data from © [Stadia Maps](#) © [Stamen Design](#) © [OpenMapTiles](#) © [OpenStreetMap](#) contributors.

The price menu and size variation, along with characteristics of study participants are shown in Table 3. On average, participants are of low socioeconomic status: the average study participant completed up to primary school and has a mean household income between 500 and 1000 cedis per month, for a mean household size of 4.57. Almost all households (over 97 percent) burn charcoal as their primary fuel. Though 64.58% of households have cooked with LPG in the past, only 24.35 percent report having used it in the last six months.

Table 3. Urban study and participant statistics

	Mean	SD	Min	Max
LPG price per kg	7.25	2.79	0.33	10.07

Distance to depot (m)	736.8	425.3	154.8	2268.70
	3	3	5	
Participant age	43.06	16.24	19	89
Household size	4.57	2.78	1	21
Has any regular income	0.95	0.22	0	1
Purchases charcoal	0.97	0.16	0	1
Used LPG in past 6 months	0.26	0.44	0	1

Notes: Summary statistics for main urban study treatments (price and distance) and study participants.

Main findings

We summarize results from both the rural and urban studies, then discuss policy implications.

Rural study

These analyses rely on household transactions at the randomly assigned LPG depot, where empty 14.5kg cylinders were exchanged for full cylinders.

LPG Demand in rural areas is very sensitive to price and distance. Figure 4 illustrates that demand is sensitive to price and distance as well as the interaction of these variables. The figure groups households into those assigned a low price (0 to 0.28 cedis per kg) and those assigned a high price (3.37 to 4.87 cedis per kg). Those assigned an intermediate price (0.28 to 3.37 cedis per kg) are excluded from this figure. The predicted mean kgs per month, as a function of distance, are plotted for each price group. The figure shows that when prices are low, assigning households to an LPG depot located further from their community decreases demand. When prices are high, on the other hand, demand is low regardless of distance. Even when households had a depot in their own community, they did not transact at the highest price. The interaction term in Table 4 shows a positive relationship, which implies that price and distance are substitutes in demand.⁴

As the price increases from the lowest level of 0 cedis per kg to the highest level of 5 cedis per kg, households go from using over 5 kilograms (kgs) of LPG per month to close to zero kgs over the time period (Figure 6, top panel). Demand is also decreasing in distance, though the patterns show more variation as distance increases (Figure X, bottom panel). These patterns can also be seen in the regression results in Table 4, where a 1 cedi increase in the price per kg results in an average 1 fewer kgs per

⁴ Table 4 breaks the result into the probability of any transactions and the number of kilograms conditional on making any purchases. The same pattern of results holds for each.

household-month, and a 10 km increase in the distance results in 3 fewer kgs per household month.

We also assess other demand determinants, most importantly whether the household (randomly) had past experience with LPG due to assignment to a previous trial arm that either provided free LPG (GRAPHs) or free delivery (ELAG). Figure 4 shows that mean monthly LPG use is 60 percent higher for households that live in a community with past LPG experience. Table 5 tests for the effect of other potential demand determinants, and shows that being located in a community randomly selected for an LPG depot increases demand. On the other hand, being located in a more convenient location (same quadrant, which proxies for transportation difficulties) does not affect demand. Finally, demand is neither increasing nor decreasing over the course of the project.

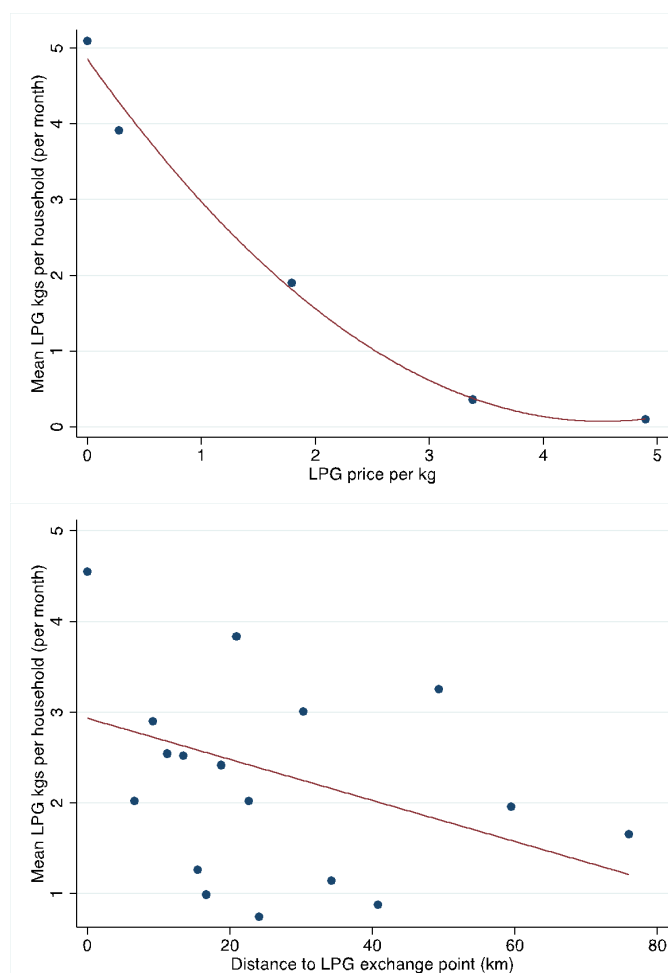


Figure 3: Rural monthly household-level LPG use by price (top) and distance (bottom)

Notes: Monthly kgs of LPG per household. The top figure shows the mean number of monthly kgs of LPG for households in each price treatment and the bottom figure shows the mean number of kgs for households in each distance treatment.

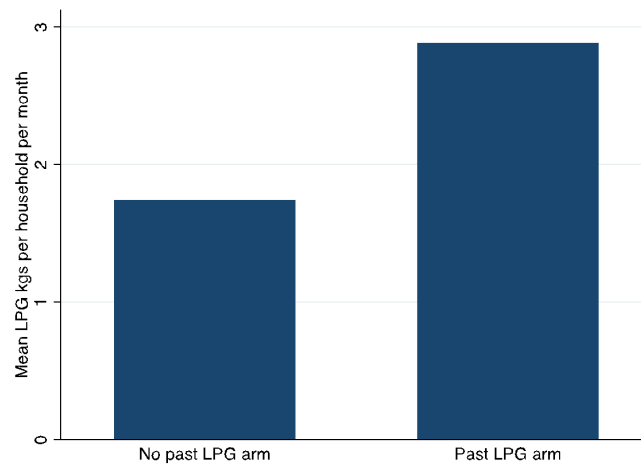


Figure 4: Probability of LPG use by LPG experience

Notes: Mean monthly household LPG purchases over the six month study period. Past LPG experience is based on random assignment to treatment in a prior intervention study and included free LPG or free delivery, through GRAPHs or ELAG, respectively.

Urban study

These analyses rely on household transactions at the randomly assigned LPG depot, where empty 3 or 14.5kg cylinders were exchanged for full cylinders.

Demand for LPG in urban settings is sensitive to price; distances were much closer than in the rural study. As the price increases from the lowest level of 0 to 2 cedis per kg to the highest level of 10 cedis per kg, households go from exchanging an average of 5 kgs per month to close to 3.2 kgs per month (Figure 5, top panel). These patterns can also be seen in the regression results in Table 7, where a 1 cedi increase in the price per kg results in an average 0.5 fewer kg per month.

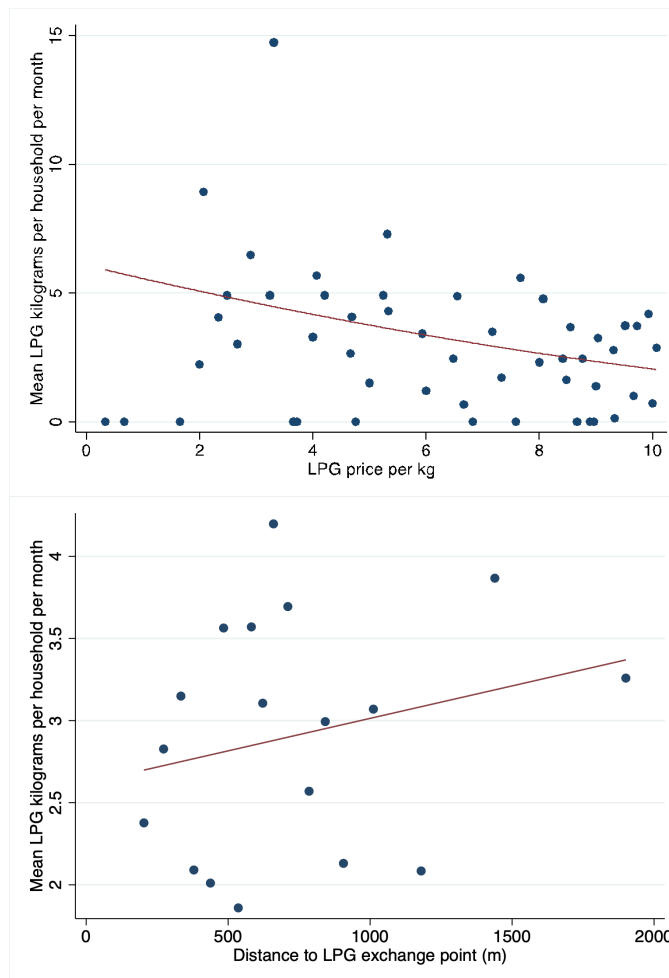


Figure 5: Urban mean monthly household-level LPG use by price (top) and distance (bottom)

Notes: Transactions per household over the three-month study window. The top figures shows the mean number of transactions for households at each price treatment and the bottom figure shows the mean number of transactions for households at each distance.

We also assess other demand determinants, most importantly whether the household had past experience with LPG. Figure 6 shows that LPG demand over the three-month study window is higher for households with past LPG experience.

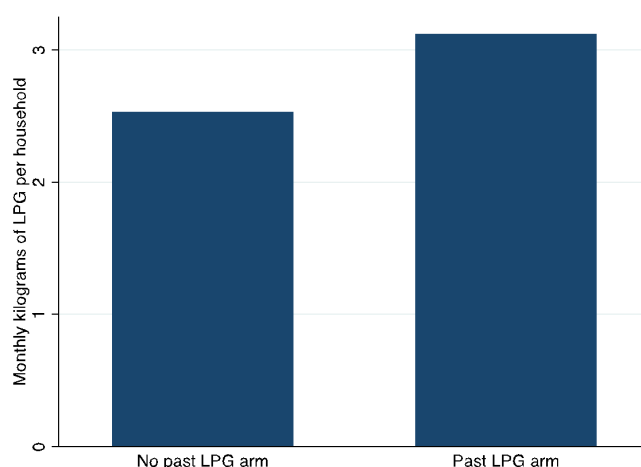


Figure 6: Urban monthly LPG use by LPG experience

Notes: Mean monthly household LPG purchases over the three-month study period based on whether or not the user had used LPG in the six months before the baseline survey was conducted.

Policy implications

We use the results from the project to draw four policy implications.

Implication 1: Demand is sensitive to price, with greater sensitivities in rural areas

Demand is decreasing in LPG price and in rural areas with distance to LPG station. The magnitudes are meaningful. For example, a rural household offered the lowest prices in the study (0.13 cedis per kg) exchanged an average of 4.67 kilograms of LPG per month. We estimate that this represents close to half of exclusive use for the average study household. On the other hand, households offered a high price (roughly 4 cedis per kg) exchanged an average of 0.33 kilograms of LPG per month, representing little change from status quo cooking practices (mostly firewood). An urban household offered the lower range of prices (greater than or equal to 4 cedis per kg) exchanged an average of 4.43 kgs of LPG during the three-month study, while those offered higher prices (greater than or equal to 8 cedis per kg) exchanged 2.94 kgs of LPG per month.

At all price levels, rural households with depots in their own communities, in other words, those for whom distance represented the lowest barrier, purchased an average of 3.8 kgs of LPG per month, while those who were assigned a depot more than 30 km away used only 1.59 kgs of LPG per month. Thus, increasing demand for LPG among rural households requires lower prices and for rural households more convenient access to LPG.

Implication 2: In rural areas, when prices are low distance to the LPG exchange point affects demand

Implication 3: At anything close to market prices, LPG take up by rural communities is effectively zero.

We discuss implications 2 and 3 together. Price and distance are substitutes in LPG demand for rural households. In other words, distance becomes less important when prices are high and more important when prices are low. This relationship is driven by the pronounced effect of price on demand: when prices are high, very few rural households will buy LPG, even when it can be purchased within their own community. On the other hand, when prices are low, increasing convenience leads to even higher demand. This is less salient in urban communities as the furthest depots were 2.3 km away.

Implication 4: Past experience with LPG helps boost demand.

Past experience with LPG leads to almost 60 percent higher LPG use in the rural study, and 22 percent higher in the urban. This may be due to learning (main cooks know how to use LPG), habit formation, or something else, and suggests that users enjoy using LPG and will take it up again when given the chance, but only if it is affordable and accessible.

Additional results

The text, figures, and tables below offer a more detailed picture of the study findings, and are referenced above in the text.

Figure 7 illustrates that demand is sensitive to price and distance as well as the interaction of these variables. The figure groups households into those assigned a low price (0 to 0.28 cedis per kg) and those assigned a high price (3.37 to 4.87 cedis per kg). Those assigned an intermediate price (0.28 to 3.37 cedis per kg) are excluded from this figure. The predicted mean kgs per month, as a function of distance, are plotted for each price group. The figure shows that when prices are low, assigning households to an LPG depot located further from their community decreases demand. When prices are high, on the other hand, demand is low regardless of distance. Even when households had a depot in their own community, they did not transact at the highest price. The interaction term in Table 4 shows a positive relationship, which implies that price and distance are substitutes in demand.⁵

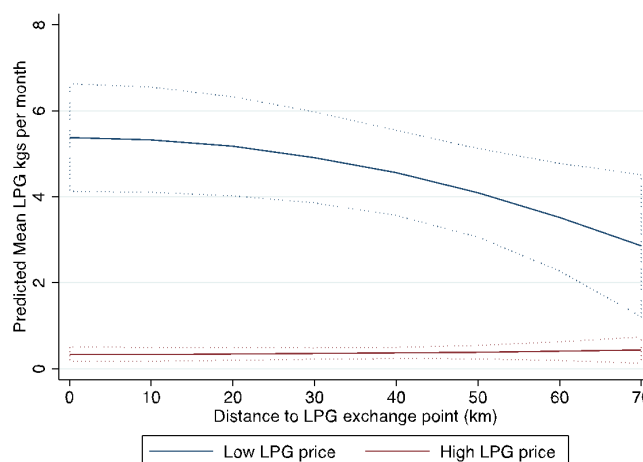


Figure 7: Monthly household-level kilograms by distance and price

Notes: Predicted demand per month, grouping price levels into low (0 and 0.28 cedis per kg) and high (3.37 and 4.89 cedis per kg) prices. The interaction of these price groups and distance to the LPG exchange point are plotted, with 95% confidence intervals shown by the dotted lines.

⁵ Table 4 breaks the result into the probability of any transactions and the number of kilograms conditional on making any purchases. The same pattern of results holds for each.

Figure 8 illustrates that demand is sensitive to price, but not to distance, although these distances are much shorter than in the rural study. We grouped households into those assigned a low price (less than or equal to 4 cedis per kg) and those assigned a high price (greater than or equal to 8 cedis per kg). Those assigned a price between 4 and 8 cedis per kg were excluded from this analysis. The predicted mean transactions per month, as a function of distance, are plotted for each price group. The figure shows that regardless of price in urban areas demand is fairly insensitive to distance. However, we see that price is the driving factor to transact even when the depot is very close to a household.

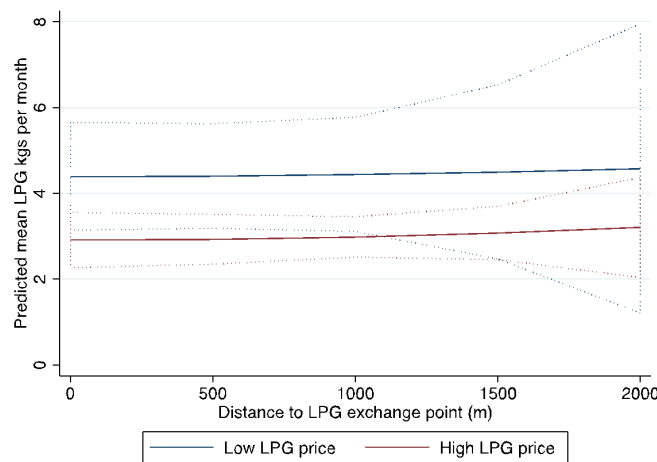


Figure 8: Monthly household-level transactions by distance and price

Notes: Predicted demand per month, grouping price levels into low (less than 4 cedis per kg) and high (more than 8 cedis per kg) prices. The interaction of these price groups and distance to the LPG exchange point are plotted, with 95% confidence intervals shown by the dotted lines.

Table 4: Rural price and distance effects on LPG demand

Number of kilograms					
Price per kilogram		-1.02*** (0.12)		-1.05*** (0.12)	-1.41*** (0.17)
Distance to depot			-0.03** (0.01)	-0.02** (0.01)	-0.05*** (0.02)
Price x Distance					0.01*** (0.00)
Constant	2.33*** (0.26)	4.45*** (0.50)	3.10*** (0.40)	5.22*** (0.57)	5.95*** (0.69)
Household-month	6,189	6,168	5,888	5,881	5,881
R-squared	0.00	0.11	0.02	0.12	0.13

Notes: Column 1 displays the constant only, which is the mean number of transactions per household. Column 2 estimates the effect of the randomized price, measured in Cedis. Column 3 estimates the effect of the randomized distance, measured in kilometers. Column 4 includes both and column 5 includes their interaction. Standard errors are clustered at the village level. No controls are included.

Table 5: Rural extensive vs intensive margin of LPG demand

	Any LPG purchase		Number of kilograms	
LPG price		-0.01*** (0.00)		-0.11*** (0.03)
Distance to depot		-0.01** (0.00)		-0.02 (0.05)
Price x Distance		0.00*** (0.00)		0.00 (0.00)
Constant	0.39*** (0.03)	0.81*** (0.06)	5.82*** (0.62)	7.61*** (1.42)
Household-month	778	741	304	303
R-squared	0.00	0.27	0.00	0.04

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: Columns 1 and 2 show the probability of any LPG purchase over the six-month study window. Columns 3 and 4 show the total household-level number of kilogram use, conditional on any purchase.

Table 6: Rural other effects on LPG demand			
	Number of kilograms		
Same quadrant	0.50 (0.37)	0.30 (0.36)	0.57* (0.33)
Past LPG arm	1.12** (0.49)	1.01** (0.48)	1.10** (0.46)
Depot in community		1.20* (0.70)	
Project Month			0.09 (0.06)
Constant	1.56*** (0.25)	1.61*** (0.24)	0.87*** (0.32)
Household-months	5,825	5,825	5,000
R-squared	0.03	0.03	0.01

Notes: Column 1 shows the effect of the stratification variables – enrollment in a past LPG arm in GRAPHS or ELAG and an indicator for being assigned to a depot in the same quadrant (road network). Column 2 adds an indicator for whether the individual is in a depot community. Column 3 includes a continuous measure of time (project month).

Table 7: Urban price and distance effects on transactions

LPG exchanged (kg per month)					
LPG price per kg		-0.47*** (0.07)		-0.47*** (0.07)	-0.56*** (0.12)
Distance to depot			0.00 (0.00)	0.00 (0.00)	-0.00 (0.00)
Price x Distance					0.00 (0.00)
Constant	0.69*** (0.03)	7.16*** (0.67)	3.45*** (0.43)	6.85*** (0.72)	7.52*** (1.07)
Households	425	425	425	425	425
R-squared	0.00	0.11	0.01	0.12	0.12

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Notes: In the first regression, we exclude delivery failures that resulted in no LPG exchanged.

Table 8: Urban extensive vs intensive margin of transactions

		Any exchange		Total exchanges	
LPG price per kg		-0.03		-0.33***	
		(0.02)		(0.06)	
Distance to depot		-0.00		-0.00	
		(0.00)		(0.00)	
Price x Distance		0.00		0.00	
		(0.00)		(0.00)	
Constant	0.54***	0.71***	2.12***	4.44***	
	(0.02)	(0.15)	(0.08)	(0.50)	
Households	543	525	292	292	
R-squared	0.00	0.02	0.00	0.17	

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1